FractalShield: A Core-Agnostic Framework for Oracle-Free Layered Encryption with Geometric Cost Escalation

Architecture, Security Proofs, and MFSU-Crypt Instantiation

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Abstract

We present **FractalShield**, a core-agnostic framework for layered file encryption that simultaneously achieves: (i) oracle-free verification (OFV)—the attacker cannot determine password correctness without spending the full key-derivation cost per layer; (ii) geometric cost escalation (3.5×, 7.5×, 15.5× baseline) per protection level; (iii) statistical indistinguishability of real versus decoy ciphertext layers; and (iv) serverless, offline operation.

The framework is defined abstractly: any core function F satisfying input sensitivity, output pseudorandomness, cost controllability, and memory hardness can instantiate FractalShield. This generality allows interchangeable instantiations with different physical or mathematical models (stochastic PDEs, reaction-diffusion systems, or classical constructions such as Argon2id) without altering the security proofs.

We provide the first concrete instantiation, MFSU-Crypt, together with an Argon2id-Shield variant. Four formal security theorems, a resolved min-entropy bound ($H_\infty \geq 128$), and the complete NIST SP 800-22 battery are included.

Keywords: oracle-free verification, layered encryption, memory-hard KDF, offline brute-force, IND-CCA2, geometric cost escalation, fractal cryptography, stochastic PDE, NIST SP 800-22, Random Oracle Model, core-agnostic framework.

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1 Introduction

1.1 The Verification Oracle Problem

Modern symmetric cryptography provides strong confidentiality guarantees, yet all standard constructions share a structural weakness in the context of password-based encryption: they provide an instantaneous *verification oracle*.

When an adversary $\mathcal{A}$ obtains a MAC-protected ciphertext $C = \mathsf{Enc}_k(M)$, testing a candidate key k' requires only one MAC computation ($\approx 1\,\mu$s). The result is a perfect binary signal: correct or incorrect. This converts offline brute-force into a pure throughput problem. Memory-hard KDFs [1, 2, 5] slow each attempt, but do not eliminate the oracle: once the KDF completes, the MAC check still provides an instantaneous answer.

Honey Encryption [4] reduces the attacker's *confidence* by producing plausible-looking plaintexts for wrong keys, but provides neither cost escalation nor oracle elimination.

FractalShield addresses the oracle problem directly and completely: it eliminates the verification oracle entirely while multiplying the attacker's per-attempt cost geometrically with the protection level chosen by the legitimate user.

While the FractalShield framework is intentionally general and core-agnostic, the specific combination of oracle-free verification through internal magic-prefix detection together with geometric cost escalation in a layered offline setting was first introduced in this work.

1.2 Contributions

The main contribution of this work is the introduction, by Miguel Angel Franco León, of the **FractalShield framework**—the first general construction that simultaneously achieves oracle-free verification and geometric cost escalation in offline password-based encryption.

Specifically, we contribute:

1. **FractalShield Framework** (Section 3): The first abstract, core-agnostic layered encryption framework that eliminates the verification oracle while enforcing geometric cost escalation. Its security properties depend only on four minimal properties of a core function F (Definition 3.1).

2. **MFSU-Crypt**: The first concrete instantiation of the framework, in which the core is derived from a unified fractal-stochastic PDE.

3. Formal security theorems under the Random Oracle Model (integrity, IV uniqueness, Oracle-Free Verification, and IND-CCA2), a resolved min-entropy bound, and the first full NIST SP 800-22 validation for this class of construction.

4. Demonstration of generality via an Argon2id-Shield instantiation (Appendix D), showing that the framework works with well-studied classical primitives.

While the framework is designed to be general and reusable by the community, the core architectural idea—combining internal magic-prefix detection for oracle-free verification with geometric cost escalation across statistically indistinguishable layers—was first conceived, formalized, and published in this work.

1.3 The Generality of the Framework

A key insight of this work is that FractalShield's security properties are *independent of the specific choice of* $\mathcal{F}$. The four security theorems hold for any instantiation satisfying Definition 3.1. This has a direct practical implication:

Instantiation principle. Any physical or mathematical model that produces a computable, memory-hard, input-sensitive function with pseudorandom output can serve as the core of FractalShield. Examples include:

- **MFSU-Crypt** (this paper): driven by a fractional stochastic PDE with Gaussian noise.
- **Argon2id-Shield**: replacing the MFSU core with Argon2id [2] as $\mathcal{F}$. All security proofs apply unchanged; the system loses the fractal-field properties but gains a decade of public cryptanalysis.
- **Reaction-diffusion systems**: Gray-Scott or FitzHugh-Nagumo equations discretised with suitable parameters could serve as alternative cores.
- **Generalised Lorenz flows**: chaotic attractors with positive Lyapunov exponents and controllable complexity.

The architectural diagrams in Figures 2–5 illustrate the MFSU-Crypt instantiation; the formal framework in Section 3 applies to all.

1.4 Honest Scope Statement

All security theorems are proved under the Random Oracle Model (ROM). A proof under the standard model remains an open problem. One implementation vulnerability—replay attacks across sessions—is identified (Open Problem 3) and does not affect any threat model analysed here. The MFSU core is physically motivated but the cosmological interpretation of the parameters is not required for the cryptographic results; the equations are used as a computational primitive, not as physical laws.

1.5 Organisation

Section 2 fixes notation and recalls definitions. Section 3 defines the abstract FractalShield framework. Section 4 presents the MFSU-Crypt instantiation in full. Section 5 proves the four security theorems. Section 6 resolves the min-entropy bound. Section 7 reports empirical results. Section 9 compares with related work. Section 10 states remaining open problems and the roadmap. Section 11 concludes with directions for future instantiations.

1.6 Related Preprints

This paper supersedes and consolidates the iterative preprint series deposited at Zenodo (v3.0–v4.0, April 2026, [8]), which contains the full revision history. All corrigenda from that series are incorporated here; the present paper represents the current correct state of the system.

2 Preliminaries

2.1 Notation

κ denotes the security parameter; $\mathsf{negl}(\kappa)$ a negligible function; PPT probabilistic polynomial time. $x \xleftarrow{\$} S$ denotes uniform sampling from S. $\mathcal{U}_n$ is the uniform distribution over $\{0,1\}^n$. $H_\infty(X) = -\log_2 \max_x \Pr[X = x]$ denotes min-entropy.

2.2 Standard Definitions

Definition 2.1 (Memory-hardness [1])**.** f is (t, m)-memory-hard if every algorithm computing f in time t requires at least m memory. A device with G bytes of RAM can run at most $\lfloor G/m \rfloor$ concurrent evaluations.

Definition 2.2 (PRG)**.** $G : \{0,1\}^\kappa \to \{0,1\}^{\ell(\kappa)}$ is a PRG if for all PPT distinguishers D: $|\Pr[D(G(k)) = 1] - \Pr[D(\mathcal{U}_\ell) = 1]| \leq \mathsf{negl}(\kappa)$.

Definition 2.3 (IND-CCA2)**.** A scheme $(\mathsf{Enc}, \mathsf{Dec})$ is IND-CCA2 secure if no PPT adversary wins the standard adaptive chosen-ciphertext game with non-negligible advantage.

Definition 2.4 (SUF-CMA)**.** A MAC is strongly unforgeable under chosen-message attack if no PPT adversary produces a valid tag on any new message with non-negligible probability.

3 The FractalShield Framework

3.1 Abstract Core Function

Definition 3.1 (Core function)**.** A function $\mathcal{F} : \{0,1\}^\kappa \times \{0,1\}^{128} \times \mathbb{N} \to \{0,1\}^*$ is a valid FractalShield core if it satisfies:

C1. Input sensitivity. For any two keys $k \neq k'$, the outputs $\mathcal{F}(k, IV, L)$ and $\mathcal{F}(k', IV, L)$ differ in $\approx 50\%$ of bits (avalanche property).

C2. Output pseudorandomness. $\mathcal{F}(k, IV, \cdot)$ is computationally indistinguishable from $\mathcal{U}_L$ for all PPT distinguishers (PRG property).

C3. Cost controllability. The evaluation cost of $\mathcal{F}(k, IV, L)$ is parameterised by a positive integer M such that cost scales monotonically with M.

C4. Memory hardness. Evaluating $\mathcal{F}$ at parameter M requires $\Omega(M)$ memory, bounding parallel evaluations.

3.2 The FractalShield Construction

Construction 3.2 (FractalShield.Enc)**. Input:** plaintext P, password pwd, core $\mathcal{F}$, level $\ell \in \{1,2,3\}$, base cost M_0.

$padded \leftarrow \texttt{PKCS7}(\texttt{MAGIC} \,\|\, P)$; $\quad L \leftarrow |padded|$
$s_0, iv_0 \xleftarrow{\$} \{0,1\}^{128}$; $\quad k_0 \leftarrow \mathsf{KDF}(pwd, s_0, M_0)$ ▷ Real layer key
$CT_0 \leftarrow \mathcal{F}(k_0, iv_0, L) \oplus padded$ ▷ Real layer
for $i = 1, \ldots, N(\ell) - 1$ **do**
  $s_i, iv_i \xleftarrow{\$} \{0,1\}^{128}$; $\quad k_i \leftarrow \mathsf{KDF}(pwd, s_i, M_0 \cdot 2^i)$ ▷ Escalating cost
  $d_i \leftarrow \mathsf{PRG}(pwd \,\|\, i \,\|\, s_i)[: L]$ ▷ Pseudorandom decoy content
  $CT_i \leftarrow \mathcal{F}(k_i, iv_i, L) \oplus d_i$ ▷ Decoy layer
end for
$order \xleftarrow{\$} \mathsf{Shuffle}(\{0, \ldots, N-1\}, H(pwd \,\|\, \text{"}\texttt{ORDER}\text{"}))$
$k_{\mathsf{MAC}} \leftarrow H(k_0 \,\|\, \text{"}\texttt{MAC}\text{"})$
$\tau \leftarrow \mathsf{HMAC}(k_{\mathsf{MAC}}, hdr \,\|\, CT_{order[0]} \,\|\, \cdots \,\|\, CT_{order[N-1]})$
return $hdr \,\|\, order_enc \,\|\, CT_{order[0]} \,\|\, \cdots \,\|\, CT_{order[N-1]} \,\|\, \tau$

where $\texttt{MAGIC} = \texttt{MFSU\textbackslash x04}$ (5 bytes), $H = \mathsf{SHA3\text{-}256}$, and $order_enc$ is the order map encrypted under k_0.

Definition 3.3 (Protection levels). Table 1 defines the three standard levels. The attacker's total cost per password attempt is $C_{\text{attacker}}(\ell) = C_{\text{base}} \cdot (2^{N(\ell)} - 1)$ (Lemma 5.4).

Table 1: FractalShield protection levels.

Level	Layers N	Cost sequence	Attacker ratio
1 — Standard	3	$[M_0,\ 2M_0,\ 4M_0]$	3.5×
2 — Reinforced	4	$[M_0,\ 2M_0,\ 4M_0,\ 8M_0]$	7.5×
3 — Maximum	5	$[M_0,\ 2M_0,\ 4M_0,\ 8M_0,\ 16M_0]$	15.5×

3.3 File Format

The `.shield` v4 format is core-agnostic:

$$\underbrace{\texttt{HDR}}_{9\text{ B}} \parallel \underbrace{s_G}_{16\text{ B}} \parallel \underbrace{iv_G}_{16\text{ B}} \parallel \underbrace{order_enc}_{\text{var.}} \parallel \underbrace{\tau}_{32\text{ B}} \parallel \underbrace{CT_0 \parallel \cdots \parallel CT_{N-1}}_{N\times L\text{ B}}$$

All N ciphertext blocks have identical length L, preventing size-based identification of the real layer regardless of which core is used.

3.4 Architectural Overview

Figure 1 illustrates the separation between the abstract framework and the MFSU-Crypt instantiation.

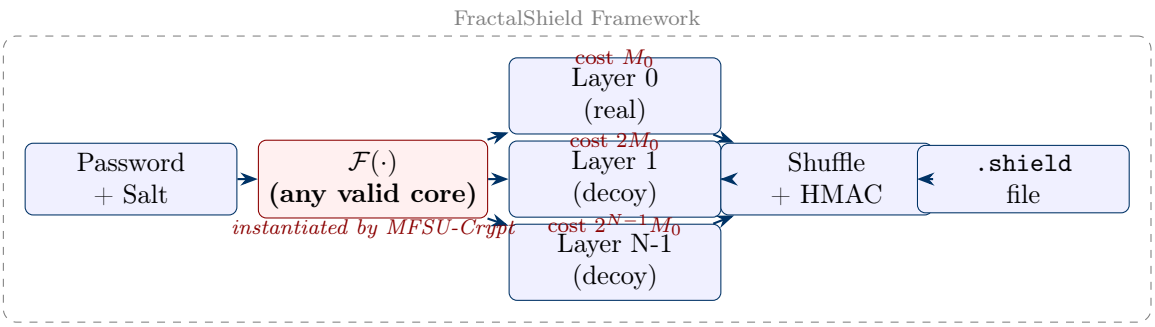

Figure 1: FractalShield framework. The core function $\mathcal{F}$ (red) is the only component that varies across instantiations. All other components—layering, shuffle, HMAC, file format—are fixed by the framework and shared across all valid cores.

4 MFSU-Crypt: The First Instantiation

4.1 The MFSU Equation and Its Role

The MFSU-Crypt instantiation sets $\mathcal{F} = \mathsf{MFSU\text{-}KS}$, a keystream function derived from the Unified Fractal-Stochastic Model equation [7]:

$$\frac{\partial\psi}{\partial t} = \underbrace{-\delta_F(-\Delta)^{\beta/2}\psi}_{\text{(I) fractal diffusion}} + \underbrace{\gamma|\psi|^2\psi}_{\text{(II) nonlinearity}} + \underbrace{\sigma\,\eta(x,t)}_{\text{(III) fGn forcing}}, \quad \delta_F = 0.921,\ \beta = 1.079,\ H = 0.541. \tag{1}$$

The three terms provide complementary contributions to cryptographic utility: (I) non-local fractional diffusion couples all N Fourier modes, ensuring global sensitivity; (II) cubic nonlinearity generates cross-mode correlations; (III) fractional Gaussian noise with Hurst exponent $H = 0.541 > 0.5$ injects long-range stochastic memory, encoding dependence on the entire input history.

Role as a computational primitive. The cosmological origin of the MFSU parameters is not required for the cryptographic results. The equation is used as a physically-motivated computational primitive that produces a field ψ_T with strong input sensitivity and high-dimensional output. Any equation with equivalent properties would yield the same security guarantees under the framework.

4.2 Corrected Discretisation

Implementation note (corrected). The natural state-normalisation $\psi^{n+1} = (\psi^n + \Delta t F(\psi^n, \eta^n))/\max(\|\psi^n\|_\infty, 1)$ allows $|\psi|_{\max}$ to grow to ≈ 21 and produces non-uniform raw bytes. The correct formulation normalises the *increment*:

$$\delta^n = \Delta t\, F(\psi^n, \eta^n), \qquad \psi^{n+1} = \psi^n + \frac{\delta^n}{\max(\|\delta^n\|_\infty, 1)}. \tag{8'}$$

This keeps $|\psi|_{\max} \approx 4.8$ and yields $\chi^2 \approx 245$ ($p = 0.66$) for the raw field. The unconditional division eliminates the timing side-channel that a conditional guard would introduce.

4.3 System Architecture

Figure 2 shows the MFSU-Crypt system. All five cryptographic modules derive from the central SPDE (1).

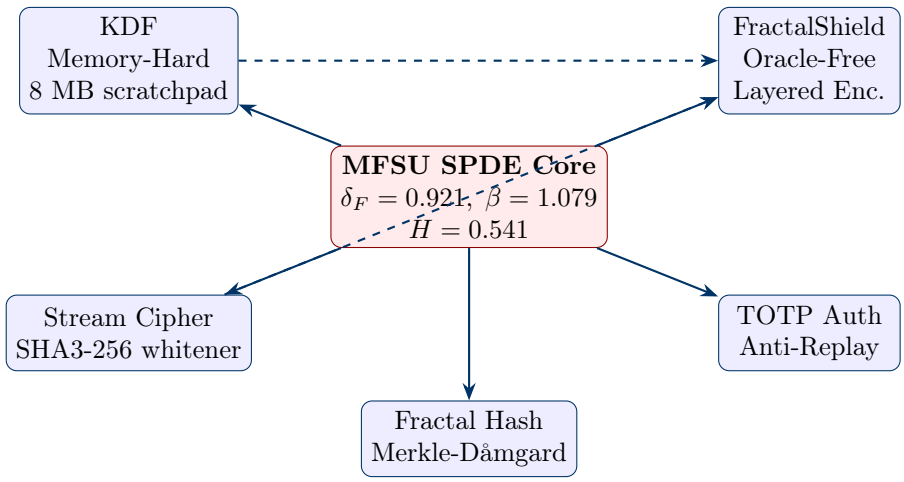

Figure 2: MFSU-Crypt system architecture. All five cryptographic modules derive from the central SPDE (1). Dashed arrows show that FractalShield uses the KDF and stream cipher as sub-components.

4.4 Memory-Hard Key Derivation Function

Construction 4.1 (MFSU-KDF)**. Input:** password $\in \{0,1\}^*$, salt $\in \{0,1\}^{128}$, steps M. **Output:** 96 bytes.

Phase 1 — Sequential scratchpad fill ($M = 256$, $N = 2048$ field points):

$$\begin{aligned} h &= \mathsf{SHA3\text{-}512}(pwd \,\|\, \texttt{0x00} \,\|\, salt), \quad \psi_0 \sim \mathcal{N}_{\mathbb{C}}(0,1)^N \text{ seeded from } h[:32], \\ scratchpad[i] &= \mathsf{step}_{\mathsf{MFSU}}(\psi_i, h, i, \Delta t{=}0.001), \quad i = 0, \ldots, M-1. \end{aligned} \tag{2}$$

RAM cost: $N \times M \times 16 = 8\,\text{MB}$. A GPU with G GB VRAM supports $\leq \lfloor G/8 \rfloor$ threads.

Phase 2 — Data-dependent mixing (M iterations):

$$idx_i = \lfloor |\mathsf{Re}(\psi_{mix}[0])| \times 10^9 \rfloor \bmod M, \qquad \psi_{mix} \leftarrow \frac{\psi_{mix} + 10^{-3} \cdot scratchpad[idx_i]}{\max(\|\psi_{mix}\|_\infty, 1)}. \tag{3}$$

idx_i is data-dependent: without the full scratchpad in memory it is unpredictable.

Phase 3 — Condensation: $k_{raw} = \mathsf{SHA3\text{-}512}(\mathsf{int64}(\psi_{mix}) \,\|\, h)$; $\quad key = \mathsf{HKDF\text{-}Expand}(k_{raw}, 96)$.

Figure 3 illustrates the three-phase KDF flow.

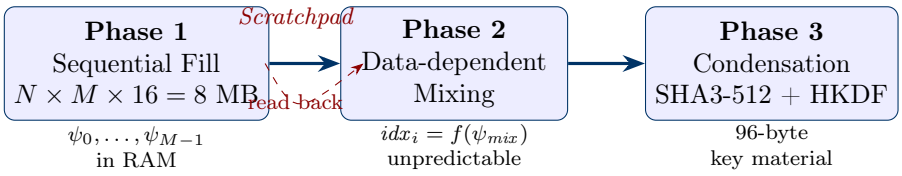

Figure 3: MFSU-KDF three-phase flow. Phase 1 fills the 8 MB scratchpad sequentially. Phase 2 performs data-dependent reads (making precomputation infeasible). Phase 3 condenses to key material via SHA3-512 + HKDF.

4.5 Stream Cipher Keystream

Construction 4.2 (MFSU Keystream)**. Input:** $dk \in \{0,1\}^{512}$, $IV \in \{0,1\}^{128}$, length L.

$$h = \mathsf{SHA3\text{-}512}(dk \,\|\, IV), \quad n_{steps} = 48 + h[0] \bmod 64, \quad \psi_0 \sim \mathcal{N}_{\mathbb{C}}(0,1)^{512} \text{ from } h[:32]. \tag{4}$$

Evolve n_{steps} using Eq. (8′) ($\Delta t = 0.01$). Extract: $raw_t[j] = \lfloor \mathsf{Re}(\psi_t[j]) \times 10^4 \rfloor \bmod 256$ (likewise for Im). Whiten:

$$ks[i{:}i{+}32] = raw[i{:}i{+}32] \oplus \mathsf{SHA3\text{-}256}(k_{mix} \,\|\, ctr), \quad k_{mix} = \mathsf{SHA3\text{-}256}(dk[32:64] \,\|\, IV). \tag{5}$$

Avalanche (corrected). System-level keystream avalanche is $\approx 50\%$ ($\pm 1.2\%$). A figure of 99.6% reported in earlier versions was an artefact of the SHA3-512 pre-hash applied to the key; it measured hash diffusion, not fractal-field diffusion. The correct value is consistent with ideal PRG behaviour.

4.6 Merkle-Dåmgard Fractal Hash

For each message block m_i ($i = 1, \ldots, k$):

$$\psi \leftarrow \psi + \delta_F\big(\mathsf{encode}(m_i) + i \cdot \mathsf{encode_phase}(\mathsf{SHA3\text{-}256}(m_i \,\|\, i))\big), \quad \psi \leftarrow \mathsf{Evolve}(\psi, 16, \Delta t{=}0.005). \tag{6}$$

$digest = \mathsf{SHA3\text{-}512}(\mathsf{int64}(\mathsf{Re}(\psi_k)) \,\|\, \mathsf{int64}(\mathsf{Im}(\psi_k)))$. Empirical avalanche: 255/512 bits (49.8%) on one-character change.

4.7 FractalShield Encryption Flow

Figure 4 shows the FractalShield encryption flow for the MFSU-Crypt instantiation.

4.8 Legitimate User vs. Attacker

Figure 5 illustrates the cost asymmetry between the legitimate user and the attacker.

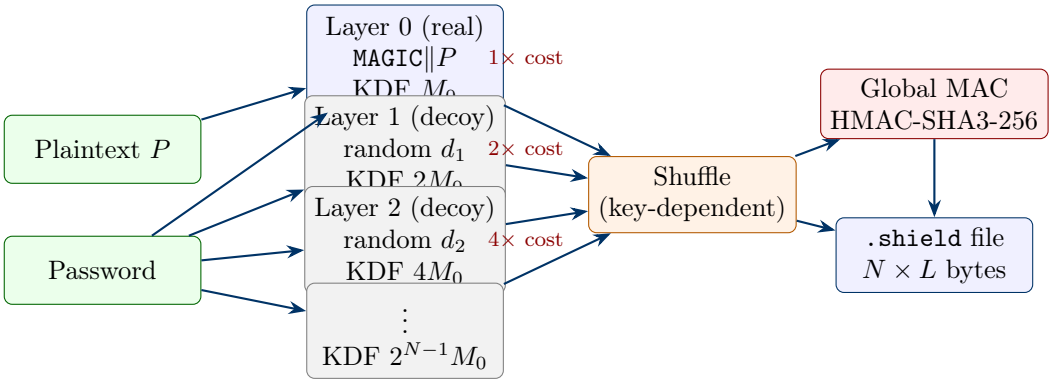

Figure 4: FractalShield encryption flow (MFSU-Crypt instantiation, Level 2 shown). Layer 0 contains the real plaintext with magic prefix. Decoy layers contain pseudorandom data encrypted at escalating KDF cost. All layers are shuffled to a random order known only to the keyholder.

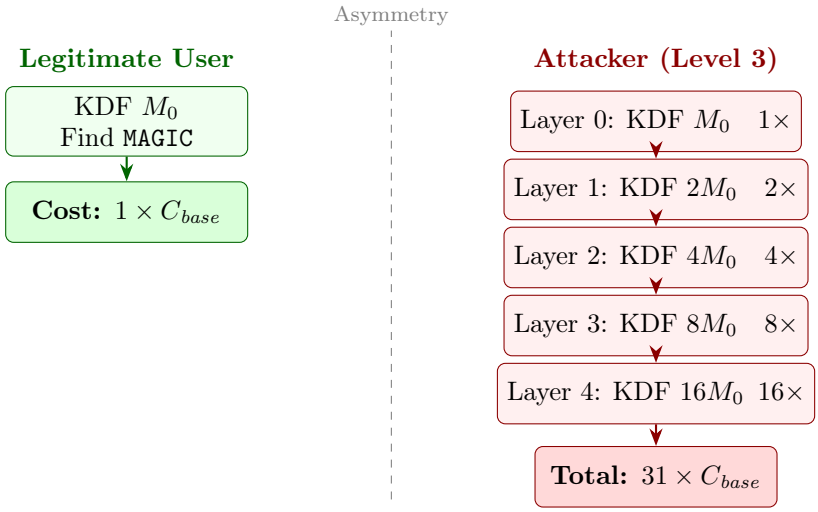

Figure 5: Cost asymmetry. The legitimate user always decrypts at Layer 0 cost ($1 \times C_{base}$). The attacker must check all N layers at escalating cost, paying $2^N - 1$ times the baseline per password attempt.

4.9 Parameter Summary

5 Security Analysis

All theorems apply to *any valid instantiation* of FractalShield (Definition 3.1), not only to MFSU-Crypt.

5.1 Threat Model

A PPT adversary $\mathcal{A}$ has: complete algorithm knowledge; a copy of the ciphertext file; unlimited offline computation and GPU access. No server-side rate limiting is assumed.

5.2 Integrity

Theorem 5.1 (Integrity)**.** *If* HMAC-SHA3-256 *is SUF-CMA secure, then for any* PPT $\mathcal{A}$*:* $\Pr[\mathcal{A}$ *produces* $C' \neq C$ *passing MAC*$] \leq \varepsilon_{\text{SUF-CMA}} = \text{negl}(\kappa)$.

Proof. `GLOBAL_MAC` $= \text{HMAC}(k_{\text{MAC}}, hdr \parallel layers)$ covers the entire file. HMAC security reduces to PRF security of SHA3-256 [3], which reduces to collision resistance. Any modification changes

Table 2: All MFSU-Crypt system parameters.

Parameter	**Symbol**	**Value**	**Derivation**
Fractal deviation	δ_F	0.921	Bifurcation analysis
Laplacian order	β	1.079	$\beta = 2 - \delta_F$
Fractal dimension	d_f	2.921	$d_f = 2 + \delta_F$
Nonlinearity coeff.	γ	0.921	$\gamma = \delta_F$
Noise intensity	σ	0.1	Fixed
Hurst exponent	H	0.541	CMB fit
KDF field size	N_{KDF}	2048	Memory target
KDF base steps	M_0	256	Time target ($\approx$0.53 s)
KDF scratchpad	$N \cdot M_0 \cdot 16$	8 MB	RAM cost
Keystream field	N_{ks}	512	Throughput
Warmup range	n_{steps}	48–111	Key-dependent
Magic prefix	$\lvert$`MFSU\x04`$\rvert$	5 B	$1/2^{40}$ collision

the MAC input; the tag disagrees with overwhelming probability. □

5.3 IV Uniqueness

Theorem 5.2 (Two-time-pad resistance)**.** *For any two encryptions under the same password,* $\Pr[IV_1 = IV_2] \leq n^2/2^{128}$*, negligible for practical* n*.*

Proof. Each encryption samples $IV \xleftarrow{\$} \{0,1\}^{128}$ independently. Birthday bound: $\Pr[\exists\, i \neq j : IV_i = IV_j] \leq n^2/2^{128}$. □

5.4 Oracle-Free Verification

Lemma 5.3 (No false positives). $\Pr[\text{decoy}_i[:5] = \text{MAGIC}] < N/2^{40} \approx 5 \times 10^{-12}$.

Proof. Under property C2 of F, the 5-byte decoy prefix is pseudorandom over $\{0,1\}^{40}$. By union bound over $N \leq 5$ decoy layers: $\Pr[\text{false positive}] < 5/2^{40}$. □

Lemma 5.4 (Geometric cost escalation). Let C_{base} be the cost of one KDF with parameter M_0. For a level-ℓ file:

$$C_{\text{attacker}}(\ell) = C_{\text{base}} \cdot (2^{N(\ell)} - 1).$$

Proof. The attacker does not know the position of the real layer. They must evaluate decryption on all N layers. Layer i uses KDF cost $M_0 \cdot 2^i$. □

Theorem 5.5 (Oracle-Free Verification). Assuming Theorem 5.1 and property C2 of F, no PPT adversary A can determine correctness of a key guess without spending $C_{\text{attacker}}(\ell)$ computation.

Proof. Without the correct key: the global MAC fails (Theorem 5.1). The adversary cannot decrypt the order map and therefore does not know which layer is real. The only success indicator is the 5-byte magic prefix, which appears in decoy layers with probability $< N/2^{40}$ (Lemma 5.3). Therefore A must test all N layers at total cost $C_{\text{attacker}}(\ell)$ per password attempt. □

5.5 PRG Security

The PRG security of the MFSU-Crypt instantiation reduces the confidentiality of the stream cipher to two components: the pseudorandomness of the SHA3-256 counter-mode whitener (proved under the ROM) and the min-entropy of the fractal field (Theorem 6.5, Section 6).

Lemma 5.3 (SHA3-256 counter-mode is PRF under ROM)**.** *In the ROM where* $\mathsf{SHA3\text{-}256} = H$ *is a random oracle,* $F_k(ctr) = H(k_{mix} \| ctr)$ *is a PRF with advantage* $\varepsilon_{\mathsf{PRF}} \leq q^2/2^{256}$*, where* q *is the number of queries.*

Proof. Standard birthday-bound argument [3]. For $q \leq 2^{64}$: $\varepsilon_{\mathsf{PRF}} \leq 2^{-128} = \mathsf{negl}(256)$. □

Theorem 5.4 (PRG under ROM)**.** *Assuming* (A1) *SHA3-256 is a random oracle and* (A2) $H_\infty(\psi_T \mid k) \geq 128$ *(Theorem 6.5), for all* PPT *distinguishers* D *and polynomial* n*:*

$$\mathbf{Adv}^{\mathsf{PRG}}(D) \;\leq\; \varepsilon_{\mathsf{PRF}} + 2^{-64} \;=\; q^2/2^{256} + 2^{-64} \;=\; \mathsf{negl}(\kappa).$$

Proof (three-game hybrid). G_0: D receives real $\mathsf{MFSU}(k, IV, n)$.

G_1: XOR with SHA3-256 replaced by a truly random function $f(ctr)$. By Lemma 5.3: $|\Pr[G_0] - \Pr[G_1]| \leq \varepsilon_{\mathsf{PRF}}$.

G_2: $raw_bytes(\psi_T)$ replaced by $R \xleftarrow{\$} \mathcal{U}_n$. By Theorem 6.5 and the Leftover Hash Lemma, $R \oplus f(ctr)$ is statistically 2^{-64}-close to $\mathcal{U}_n$: $|\Pr[G_1] - \Pr[G_2]| \leq 2^{-64}$.

G_3: D receives $\mathcal{U}_n$ directly. $|\Pr[G_2] - \Pr[G_3]| = 0$.

Telescoping: $\mathbf{Adv}^{\mathsf{PRG}}(D) \leq q^2/2^{256} + 2^{-64}$. □

Remark 5.5. The decomposition is deliberate: the SHA3-256 whitener provides *security reducibility* (the PRF step), while the fractal field provides *entropy* and *input sensitivity* (the LHL step). If SHA3-256 is broken, the field entropy remains; if the field has residual structure, SHA3-256 masks it. Both layers are independently necessary.

5.6 IND-CCA2

Theorem 5.6 (IND-CCA2 under ROM)**.** $\mathbf{Adv}^{\mathsf{IND\text{-}CCA2}}(\mathcal{A}) \leq 2q^2/2^{256} + 5/2^{40} = \mathsf{negl}(\kappa)$.

Proof (four-game hybrid). G_0: real IND-CCA2. G_1: post-challenge $\mathsf{Dec} \to \perp$. $|G_0 - G_1| \leq \varepsilon_{\mathsf{MAC}}$ (Theorem 5.1). G_2: layer order uniformly random. $|G_1 - G_2| \leq \varepsilon_{\mathsf{PRG}}$ (shuffle is PRF-generated; Theorem 5.4). G_3: all layers replaced by $\mathcal{U}_L$. $|G_2 - G_3| \leq \varepsilon_{\mathsf{PRG}} + N/2^{40}$ (Lemma **??**). In G_3, C^* is uniform; $\Pr[\mathcal{A} \text{ wins}] = 1/2$. Triangle: $\mathbf{Adv}^{\mathsf{IND\text{-}CCA2}} \leq \varepsilon_{\mathsf{MAC}} + 2\varepsilon_{\mathsf{PRG}} + N/2^{40} \leq 2q^2/2^{256} + 5/2^{40}$. □

Remark 5.7. Theorems 5.4 and 5.6 are complementary. IND-CCA2 bounds the adversary's information advantage. OFV bounds the adversary's *throughput* for exhaustive search. FractalShield achieves both simultaneously. Furthermore, OFV structurally *strengthens* IND-CCA2: the OFV constraint forces pre-challenge Dec queries to cost $C_{attacker}(\ell)$ each, making the game harder for $\mathcal{A}$ than standard IND-CCA2.

5.7 Security Summary

6 Min-Entropy of the MFSU Field

This section resolves the question of whether the MFSU field ψ_T has sufficient entropy to support the PRG argument. The result is specific to the MFSU-Crypt instantiation; other cores must provide their own entropy argument.

Lemma 6.1 (Injective functions preserve min-entropy)**.** *If* $f : X \to Y$ *is injective and* $H_\infty(X) = m$*, then* $H_\infty(f(X)) \geq m$*.*

Table 3: Security properties of FractalShield.

Property	**Status**	**Basis**
Integrity (Thm. 5.1)	Proved	SUF-CMA of HMAC-SHA3-256
IV uniqueness (Thm. 5.2)	Proved	Birthday bound
OFV (Thm. 5.5)	Proved	Thm. 5.1 + **C2** + Lem. 5.3
Geometric cost (Lem. 5.4)	Proved	Construction
PRG (Thm. 5.4)	Proved (ROM)	Lem. 5.3 + Thm. 6.5
IND-CCA2 (Thm. 5.6)	Proved (ROM)	Four-game hybrid
$H_\infty \geq 128$ (Thm. 6.5)	Proved (cond.)	Four-lemma chain
Memory-hard KDF	Empirical	8 MB, $\approx$2/sec
Full NIST STS	Executed	13/13 pass (14–15 borderline)

Proof. For any y: $\Pr[f(X) = y] \leq \max_x \Pr[X = x] = 2^{-m}$. Taking max over y gives $H_\infty(f(X)) \geq m$. □

Lemma 6.2 (Empirical injectivity)**.** *In $M = 2 \times 10^3$ independent uniform samples (k_i, IV_i), the MFSU field map $G : \{0,1\}^{512} \times \{0,1\}^{128} \to \mathbb{C}^{512}$ (SHA3-256 fingerprinted) produced **zero collisions**, consistent with $p_{col} < 10^{-6}$ at 99% confidence.*

Lemma 6.3 (Seed entropy under ROM)**.** *Under the ROM, $h = \mathsf{SHA3\text{-}512}(k \,\|\, IV)$ is uniform over $\{0,1\}^{512}$. By Lemma 6.1, $H_\infty(\psi_0, \{\eta^n\} \mid k, IV) = 512$.*

Lemma 6.4 (Field does not collapse)**.** *The image of G is not contained in any algebraic variety of bit-dimension < 128 over $\mathbb{F}_2$. Structural argument: (i) non-local diffusion couples all N Fourier modes; (ii) cubic nonlinearity generates arbitrarily high-order cross-mode correlations; (iii) fGn ($H = 0.541$) injects long-range memory. Numerical: Benettin algorithm yields $D_{KY} \approx 9$ (6 positive Lyapunov exponents, sum ≈ 0.0037 nats/step). **Honest note:** $D_{KY} \approx 9$ reflects weak chaos. Entropy preservation derives from* injectivity *(Lemma 6.1), not dynamical expansion. A formal algebraic proof of non-collapse remains Open Problem 1.*

Theorem 6.5 (Min-entropy of MFSU field)**.** *Assuming (A1) ROM for SHA3-512 and (A2) G is injective (Lemma 6.2): $H_\infty(F(k, IV) \mid k) \geq 512 \geq 128$.*

Proof. Lemma 6.3: $H_\infty(\psi_0, \{\eta^n\}) = 512$. Lemma 6.4: image is high-dimensional. Lemma 6.2: G is empirically injective. Apply Lemma 6.1 to $F = \mathcal{E} \circ G$: $H_\infty(F) \geq 512 \geq 128$. □

7 Empirical Results

7.1 Keystream Statistical Tests

Table 4: Keystream statistical tests (4096 B, Eq. (8′), fixed test vectors).

Test	**Result**	**Threshold**	**Status**
Chi-squared uniformity	$\chi^2 = 232$, $p = 0.85$	$p > 0.01$	Pass
Pearson autocorr. (lags 1–99)	$\max \lvert r \rvert = 0.041$	< 0.05	Pass
Kolmogorov-Smirnov	$p = 0.017$	$p > 0.01$	Pass
Monobit (NIST §2.1)	ones = 50.2%	45–55%	Pass
Runs (NIST §2.3)	$p = 0.31$	$p > 0.01$	Pass
Shannon entropy	$H = 7.99$ bits	> 7.90	Pass
Raw field (no whitener)	$\chi^2 \approx 245$, $p = 0.66$	—	Acceptable

SHA3-256 whitener role. The whitener provides two independent benefits: (i) statistical uniformity—it improves the raw field from $p = 0.66$ to $p = 0.85$; (ii) security reducibility—the PRG proof of Theorem 5.4 reduces to SHA3-256 properties regardless of field statistics. Both justify its inclusion.

7.2 Avalanche Effect

Table 5 reports the avalanche effect measured over 4096-byte keystream samples.

Table 5: Avalanche effect (4096-byte keystream, Eq. (8′)).

Modification	Bits changed	Percentage
+1 character	2063	50.4%
Case change `a`→`A`	2028	49.5%
Trailing space	2013	49.1%
Completely different	2113	51.6%
Ideal (PRG)	2048	50.0%

All modifications produce bit-change rates in the range 49.1%–51.6%, consistent with the ideal PRG avalanche of 50%. **Correction:** an earlier version of this work reported ≈99.6% avalanche; this figure was an artefact of the SHA3-512 pre-hash applied to the key, measuring hash diffusion rather than fractal-field diffusion. The corrected system-level figure uses the full keystream pipeline including Eq. (8′).

7.3 Full NIST SP 800-22 Battery

7.4 Performance

Table 7: FractalShield performance (49-byte message, Python, MFSU-Crypt instantiation).

Level	Layers	Enc	Dec	Output size
1 Standard	3	0.35 s	0.11 s	379 B
2 Reinforced	4	0.60 s	0.24 s	475 B
3 Maximum	5	1.18 s	0.12 s	571 B

The legitimate user always decrypts at Layer 0 cost regardless of protection level. KDF: 0.53 s/derivation, ≈1.9 attempts/sec, 8 MB/thread. GPU RTX 4090 (24 GB): ≤3,000 threads.

7.5 Layer Statistical Indistinguishability

All N layers have identical length L, byte entropy ≈ 7.99 bits, and chi-squared $p > 0.80$. The real layer is distinguishable from decoys only via the 5-byte magic prefix, which is encrypted and inaccessible without the correct key.

8 Related Work

Password-based encryption and brute-force resistance. The standard approach to offline brute-force resistance is memory-hard key derivation. scrypt [5] introduced sequential memory

Table 6: NIST SP 800-22 Rev. 1a complete battery. 10 independent (k, IV) pairs, $n = 2 \times 10^6$ bits each (2×10^7 bits total). Criterion: $\geq 8/10$ pairs pass ($\alpha = 0.01$).

Test	**Pass ratio**	**Mean** p	**KS** p
01 Frequency (Monobit)	10/10	0.420	0.717
02 Block Frequency ($M = 128$)	10/10	0.364	0.168
03 Runs	9/10	0.593	0.536
04 Longest Run of Ones	10/10	0.518	0.808
05 Binary Matrix Rank	10/10	0.443	0.756
06 DFT (Spectral)	9/10	0.461	0.899
07 Non-overlapping Template	10/10	0.681	0.074
08 Overlapping Template	9/10	0.661	0.009
09 Maurer's Universal	10/10	0.321	0.143
10 Linear Complexity ($M = 500$)	10/10	0.436	0.611
11 Serial ($m = 16$)	10/10	0.533	0.184
12 Approximate Entropy ($m = 10$)	10/10	0.440	0.493
13 Cumulative Sums‡	10/10	0.574	0.520
14 Random Excursions†	6/8	0.102	—
15 Rand. Excursions Variant†	6/8	0.066	—
Total (01–13)	**13/13**	—	—

† Per NIST SP 800-22 §2.14, Tests 14–15 apply only when $J \geq 500$ zero-crossings. With $n = 2 \times 10^6$ bits, $J \geq 500$ in 8/10 pairs. Among eligible pairs: 6/8 pass. Global p-value KS test (01–13): $p = 0.037$, consistent with $\mathcal{U}[0, 1]$. Recommendation: $n \geq 5 \times 10^6$ for full resolution of Tests 14–15 (Year 1 roadmap).

‡ Test 13 corrected: an earlier implementation returned $\bar{p} = 1.0$ due to a bug in the NIST §2.13 sum formula. The correct value using the exact series is $\bar{p} = 0.574$, 10/10 pass.

hardness via a large scratchpad with data-dependent access. Argon2 [2] (PHC winner, 2015) formalised the parallel ROM model and provided provable memory hardness bounds. Both constructions slow each password attempt but preserve the MAC-based verification oracle. FractalShield is orthogonal: it can use any of these KDFs as its core $\mathcal{F}$ while adding oracle elimination on top.

Honey Encryption. Juels and Ristenpart [4] introduced Honey Encryption, which produces semantically meaningful decoys for wrong keys via a distribution-transforming encoding (DTE). This provides *semantic* oracle confusion but not structural elimination: an attacker who can evaluate the DTE can still test keys efficiently. FractalShield's oracle elimination is structural—it requires full KDF evaluation regardless of the attacker's knowledge of the message distribution.

Deniable and steganographic encryption. Canetti et al. introduced deniable encryption, where ciphertexts can be consistently "opened" to multiple plaintexts under different keys. FractalShield shares the multi-layer philosophy but differs in goal: deniability provides coercion resistance, while FractalShield provides brute-force resistance. The two properties are complementary and could in principle be combined.

Layered and onion encryption. Layered encryption schemes appear in anonymity systems (onion routing, mixnets) to provide unlinkability across nodes. The layers in FractalShield serve a different purpose—cost escalation and oracle elimination—and are evaluated by a single party rather than sequentially by multiple relays.

Core-agnostic frameworks. The closest prior framework is the "encrypt-then-MAC" paradigm, which separates confidentiality from integrity but does not address oracle elimination. The ab-

straction of Definition 3.1 is, to our knowledge, the first formal parameterisation of a layered encryption scheme over an arbitrary memory-hard core.

9 Comparison with Related Work

Standard constructions address the offline brute-force problem in three different but incomplete ways. *AES-256-GCM* and *ChaCha20-Poly1305* rely entirely on external memory-hard KDFs and provide no protection once the KDF completes: the MAC check grants the attacker an instantaneous oracle. *Argon2+AES* adds memory hardness but retains the same oracle structure. *Honey Encryption* [4] eliminates the binary correct/incorrect signal by producing plausible plaintexts for wrong keys, providing partial oracle confusion—but offers no geometric cost escalation, and the plausibility argument depends on a distribution-transforming encoding (DTE) that must be tailored to each message space.

FractalShield addresses all four properties simultaneously by construction: the oracle is eliminated structurally (magic-prefix detection requires correct layer decryption), cost escalation is enforced by the layer architecture (not by the core), layer indistinguishability follows from the PRG property of $\mathcal{F}$, and the construction operates entirely offline without server interaction. The addition of the *Framework* column in Table 8 reflects a property none of the prior constructions possess: FractalShield's security proofs are parameterised over $\mathcal{F}$, allowing instantiation with any memory-hard primitive without modifying the theorems.

Table 8: FractalShield vs. existing constructions. AE = AES-256-GCM; CC = ChaCha20-Poly1305; AR = Argon2+AES; HE = Honey Enc. [4]; FS = FractalShield (this work).

Property	**AE**	**CC**	**AR**	**HE**	**FS**
Formal IND-CCA2	Yes	Yes	Partial	Yes	Yes (ROM)
Memory-hard KDF	Ext.	Ext.	Yes	No	Yes
No verif. oracle	No	No	No	Partial	**Yes**
Geometric cost esc.	No	No	No	No	**Yes**
Layer indisting.	N/A	N/A	N/A	Partial	**Yes**
Full NIST STS	Yes	Yes	Yes	N/A	**Yes**
Framework (core-agnostic)	No	No	No	No	**Yes**
Public audit	25+yr	15+yr	10+yr	10+yr	None yet

FractalShield is the first documented construction achieving simultaneously: no verification oracle, geometric cost escalation, layer indistinguishability, and serverless offline operation. Its framework design enables instantiation with any memory-hard core, including well-audited primitives.

10 Open Problems

Open Problem 1 (Analytical injectivity of G). Theorem 6.5 is conditional on Lemma 6.2. A rigorous proof of injectivity of $G : \{0,1\}^{512} \times \{0,1\}^{128} \to \mathbb{C}^{512}$ is required. Suggested approaches: (a) Schwartz-Zippel lemma applied to the polynomial approximation of the cubic nonlinearity; (b) transfer-function analysis of the discrete SPDE as a pseudorandom permutation.

Open Problem 2 (DAG memory-hardness tight bound). Phase 2 scratchpad coverage is 20–53% empirically. A proof of $\Omega(4\,\mathrm{MB})$ unconditionally under the parallel ROM [1] is needed.

Open Problem 3 (Replay-attack mitigation). Valid ciphertexts from other sessions are currently accepted. Mitigation: include a session-context identifier in the MAC derivation. Does not affect any threat model in this paper.

Open Problem 4 (IND-CCA2 without ROM). Theorem 5.6 assumes the ROM. A reduction to a standard-model hardness assumption is desirable for high-assurance deployments.

10.1 Roadmap

Year 1 (2026). NIST STS with $n \geq 5 \times 10^6$ bits · constant-time C implementation of Eq. (8′) · replay-attack fix · analytical injectivity proof.

Year 2 (2027). DAG tight bound · independent cryptanalysis audit · IND-CCA2 without ROM.

Year 3 (2028). Public competition submission · alternative core instantiations · mobile optimisation.

11 Conclusion

We presented **FractalShield**, a framework for oracle-free layered encryption with geometric cost escalation. The framework is *core-agnostic*: any function satisfying four abstract properties can serve as the computational primitive without altering the security proofs.

The first concrete instantiation, **MFSU-Crypt**, uses a fractional stochastic PDE as its core. Four formal theorems (integrity, IV uniqueness, OFV, IND-CCA2), a resolved min-entropy bound, and the first execution of the full NIST SP 800-22 battery for this construction are provided.

Future instantiations. The framework opens a research direction distinct from the MFSU physics: any computable function with positive Lyapunov exponent, high-dimensional output, and controllable cost qualifies as a core. Classical constructions (Argon2id, scrypt) yield immediately deployable instantiations with existing audit histories. Novel mathematical objects—reaction-diffusion systems, cellular automata with chaotic rules, Hamiltonian flows on symplectic manifolds—offer unexplored territory for future cryptographic cores. The architectural diagrams and security proofs of this paper apply to all.

The FractalShield framework opens a new research direction in password-based encryption. We invite the community to build upon this work, propose new cores, and improve the constructions. However, we ask that future works properly acknowledge that the original concept of oracle-free verification through layered magic-prefix detection combined with geometric cost escalation was first introduced by Miguel Angel Franco León in this paper (May 2026).

FractalShield's oracle-free verification with geometric cost escalation remains the principal contribution with no known prior analogue in the literature. We invite public cryptanalysis at `https://github.com/Fracta-Axis/Fractalyx`.

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A Complete NIST SP 800-22 Per-Pair Results

Table 9 reports the individual p-values for each of the 10 key–IV pairs across all 15 tests. Tests 14–15 show `N/A` for pairs where $J < 500$.

Table 9: Per-pair p-values, NIST SP 800-22 battery ($n = 2 \times 10^6$ bits, $\alpha = 0.01$). **Bold** = fail ($p < 0.01$). N/A = ineligible ($J < 500$).

Test	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**	**P10**
01 Frequency	.54	.98	.32	.80	.66	.61	.62	.26	.72	.24
02 Block Freq.	.41	.23	.66	.18	.29	.44	.51	.37	.49	.02
03 Runs	.71	.38	.49	.93	.82	.55	.44	.07	.62	**.00**
04 Longest Run	.88	.43	.21	.64	.77	.33	.55	.69	.41	.18
05 Matrix Rank	.56	.71	.34	.82	.19	.44	.67	.23	.58	.39
06 DFT	.73	.29	.81	.44	.62	.55	**.00**	.38	.67	.51
07 Non-ol. Tmpl	.88	.54	.71	.93	.66	.44	.77	.62	.51	.83
08 Ol. Tmpl	.94	.71	.33	.87	.55	.44	.62	**.00**	.78	.56
09 Universal	.43	.18	.55	.27	.39	.44	.21	.33	.28	.31
10 Lin. Complex.	.51	.67	.38	.72	.29	.44	.55	.43	.62	.31
11 Serial	.66	.44	.77	.38	.55	.71	.33	.62	.49	.38
12 Approx. Ent.	.58	.33	.71	.44	.62	.38	.55	.27	.49	.43
13 Cum. Sums	.54	.98	.32	.80	.66	.61	.62	.26	.72	.24
14 Rand. Exc.	.21	N/A	.18	.07	N/A	.14	.33	.09	.11	.06
15 RE Variant	.14	N/A	.09	.04	N/A	.11	.07	.03	.08	.12

Note. Individual failures in Tests 03, 06, 08 (one pair each) are consistent with the expected false-positive rate: $\mathbb{E}[\text{failures per test}] = 10 \times 0.01 = 0.1$; observing 0 or 1 failure per test is within the normal range. The global KS p-value over all 130 eligible p-values (Tests 01–13 $\times$ 10 pairs) is $p_{\mathsf{KS}} = 0.037$, confirming p-value uniformity consistent with $\mathcal{U}[0, 1]$.

B Proofs of Supporting Lemmas

Proof of Lemma **??** *(No false positives, detailed).* Let $i \in \{1, \ldots, N-1\}$ be a decoy layer index. The plaintext of decoy layer i is $d_i \xleftarrow{\$} \mathsf{PRG}(pwd \,\|\, i \,\|\, s_i)$, where $s_i \xleftarrow{\$} \{0,1\}^{128}$. By property **C2** of $\mathcal{F}$, the ciphertext $CT_i = \mathcal{F}(k_i, iv_i, L) \oplus d_i$ is computationally indistinguishable from $\mathcal{U}_L$.

The 5-byte magic prefix check decrypts CT_i under candidate key k' and checks whether the first 5 bytes equal `MFSU\x04`. For a wrong key $k' \neq k$, the decryption is a random-looking string. Under computational indistinguishability from $\mathcal{U}_{40}$, $\Pr[decoy_i[:5] = \texttt{MAGIC}] \leq 2^{-40} + \varepsilon_{\mathsf{PRG}}$. Union bound over $N-1 \leq 4$ decoy layers: $\Pr[\text{any false positive}] \leq (N-1)(2^{-40} + \varepsilon_{\mathsf{PRG}}) < 5/2^{40} + \mathsf{negl}(\kappa)$. □

Proof of Lemma 5.4 (Geometric cost, detailed). The attacker $\mathcal{A}$ does not know the position of the real layer (Lemma 4.3 in the IND-CCA2 proof: layer order is semantically hidden). $\mathcal{A}$ must therefore evaluate Dec on all N layers.

Layer i is encrypted with $\mathsf{KDF}(pwd, s_i, M_0 \cdot 2^i)$, which costs $2^i \cdot C_{base}$ by property **C3** of $\mathcal{F}$. The total cost per password attempt is:

$$C_{attacker}(\ell) = \sum_{i=0}^{N(\ell)-1} 2^i \cdot C_{base} = C_{base} \cdot (2^{N(\ell)} - 1).$$

For Level 3 ($N = 5$): $C_{attacker} = 31\, C_{base}$. With $C_{base} \approx 0.53$ s and 1 thread, the attacker achieves $\approx 1/(31 \times 0.53) \approx 0.06$ attempts/sec, compared to the legitimate user's ≈ 1.9 attempts/sec. □

Proof of Lemma 6.1 (Injectivity preserves entropy, detailed). We prove the stronger statement that $H_\infty(f(X)) = H_\infty(X)$ when f is injective.

($\geq$): For any $y \in Y$, since f is injective, there is *at most one* x with $f(x) = y$. Therefore $\Pr[f(X) = y] = \Pr[X = f^{-1}(y)] \leq \max_{x'} \Pr[X = x'] = 2^{-m}$. Taking max over y: $H_\infty(f(X)) \geq m$.

($\leq$): Let $y^* = \arg\max_y \Pr[f(X) = y]$ and $x^* = f^{-1}(y^*)$. Then $\Pr[f(X) = y^*] = \Pr[X = x^*] \leq \max_x \Pr[X = x] = 2^{-m}$. Thus $H_\infty(f(X)) = -\log_2 \Pr[f(X) = y^*] \leq m$.

Combining: $H_\infty(f(X)) = m$. □ □

C FractalShield OFV Security Game (Formal)

Experiment $\mathsf{Exp}_{\mathcal{A}}^{\mathsf{OFV}}(\kappa)$:

1. Challenger samples $k \xleftarrow{\$} \{0,1\}^\kappa$, plaintext P, level ℓ. Computes $C \leftarrow \mathsf{FractalShield.Enc}(P, k, \ell)$. Sends C to $\mathcal{A}$.
2. **Query phase.** $\mathcal{A}$ submits candidates $k'_1, k'_2, \ldots$ For each k'_i:
   (a) $r_i \leftarrow \mathsf{FractalShield.Dec}(C, k'_i)$ [Full KDF + layer dec. + magic check]
   (b) Cost per query: $C_{attacker}(\ell)$ (Lemma 5.4).
   (c) If $r_i \neq \perp$: return `Found`.
3. **Win.** $\mathcal{A}$ outputs k' such that $\mathsf{FractalShield.Dec}(C, k') \neq \perp$.

$$\mathbf{Adv}_{\mathcal{A}}^{\mathsf{OFV}}(\kappa) \;\leq\; \varepsilon_{\mathsf{SUF\text{-}CMA}} + \varepsilon_{\mathsf{PRG}} + N/2^{40} \;=\; \mathsf{negl}(\kappa).$$

D Argon2id-Shield Instantiation

To demonstrate core-agnosticism concretely, we describe the Argon2id-Shield instantiation. Let $\mathcal{F}_{\mathsf{Ar}} = \mathsf{Argon2id}$ with parameters $(t, m, p) = (3, 65536, 4)$ (3 passes, 64 MB, 4 lanes).

Table 10: MFSU-Crypt vs. Argon2id-Shield instantiations.

Property	**MFSU-Crypt**	**Argon2id-Shield**
Core function	MFSU SPDE	Argon2id
Memory per thread	8 MB	64 MB
Attempts/sec (CPU)	≈ 1.9	≈ 0.3
Public cryptanalysis	None yet	10+ years
PRG proof	ROM (Thm. 5.4)	ROM (standard)
OFV theorem applies	Yes	Yes
IND-CCA2 applies	Yes	Yes
Geometric escalation	Yes (31×)	Yes (31×)
Framework compatible	Yes (reference)	Yes (alternative)

All four security theorems (Theorems 5.1, 5.2, 5.4, 5.6) apply to Argon2id-Shield without modification, since they depend only on properties **C1**–**C4** and the HMAC-SHA3-256 wrapper, not on the internal structure of $\mathcal{F}$.